

H524 Introduction to Biostatistics Project Preparation Homework GROUP ASSIGNMENT Fall 2025

Due Date: End of Week 8

Total Points: 100

Format: GROUP SUBMISSION - One PDF per group via Canvas

Group Size: 4-5 students

Overview

This assignment marks the beginning of your group final project for H524. By now, you have learned descriptive statistics, probability, confidence intervals, hypothesis testing (t-tests, ANOVA), and nonparametric methods. This assignment asks you to:

1. Form a research team of 4-5 students
2. Select a biostatistics dataset of interest
3. Develop testable research questions
4. Conduct preliminary data exploration
5. Create an analysis plan using methods from weeks 1-8

This preparation will serve as the foundation for your final project presentation and report at the end of the term.

Learning Objectives

By completing this assignment, you will:

- Practice collaborative research planning and teamwork
- Apply course concepts to real biomedical/public health data
- Develop testable hypotheses appropriate for your dataset
- Conduct comprehensive exploratory data analysis
- Create reproducible, well-documented R code
- Communicate statistical findings clearly

Timeline and Checkpoints

Week	Milestone
Week 7	Form groups; begin dataset exploration
Week 7.5	Optional: Dataset approval office hours
Week 8	Complete data exploration and analysis plan
Week 8 (end)	SUBMISSION DEADLINE
Weeks 9-10	Continue analysis (after learning new methods)
Week 10 (end)	Final project presentations and reports due

1 Part A: Group Formation and Organization (15 points)

1.1 Required Information

Submit the following information about your research team:

1.1.1 Team Members (5 points)

Complete the following table:

Name	Email	Primary Role(s)

Roles (students may hold multiple roles):

- **Project Coordinator:** Schedules meetings, tracks deadlines, ensures all deliverables are complete
- **Data Analyst:** Leads R coding, data cleaning, statistical analysis
- **Writer/Editor:** Coordinates writing of reports, ensures clarity and grammar
- **Methods Specialist:** Ensures appropriate statistical methods are used correctly
- **Presenter:** Leads development of final presentation materials

1.1.2 Communication and Collaboration Plan (5 points)

Describe your group's plan for working together:

- Primary communication method:** (e.g., email, Slack, Discord, text)
- Meeting schedule:** How often will you meet? When?
- File sharing:** How will you share code and documents? (e.g., Google Drive, GitHub, Canvas)
- Conflict resolution:** How will your group handle disagreements or unequal contributions?

1.1.3 Individual Contribution Statement (5 points)

At the end of your submission, include a brief statement from EACH group member describing:

- What you personally contributed to this assignment
- Approximately what percentage of the work you completed
- Any challenges your group faced in collaboration

Note: All group members should contribute meaningfully. If there are significant disparities in effort, please discuss with the instructor.

2 Part B: Dataset Selection and Description (20 points)

2.1 Dataset Selection

Your group must select ONE dataset for analysis. You have two options:

2.1.1 Option 1: Choose from Curated Datasets (Recommended)

The instructor has prepared several high-quality biostatistics datasets appropriate for this course:

1. **Framingham Heart Study (subset)**: Cardiovascular disease risk factors
2. **NHANES (subset)**: National health and nutrition data
3. **COVID-19 Clinical Outcomes**: Patient characteristics and outcomes
4. **Birth Weight Study**: Factors affecting infant birth weight
5. **Cancer Survival Data**: Treatment outcomes and prognostic factors
6. **Mental Health Survey**: Depression, anxiety, and demographic factors

Detailed dataset descriptions and download links will be provided on Canvas.

2.1.2 Option 2: Propose Your Own Dataset (Requires Approval)

If your group wants to use a different dataset, you must:

- Submit a 1-page justification by **Week 7, Wednesday**
- Include: dataset source, variables available, number of observations
- Explain why this dataset is suitable for H524 methods
- Receive instructor approval before proceeding

Dataset Requirements (ANY dataset must meet these criteria):

- At least 100 observations
- At least 8-10 variables (mix of continuous and categorical)
- Publicly available OR you have permission to use it
- Relevant to biostatistics, public health, or medicine
- Suitable for methods covered in weeks 1-8 (and weeks 9-10)
- Complete data documentation available

2.2 Dataset Description (20 points)

Provide a comprehensive description of your chosen dataset:

2.2.1 Source and Context (5 points)

- Full citation for the dataset (APA format)
- Brief background: What was the original study purpose?
- Population studied (who, where, when)
- Study design (cross-sectional, cohort, experimental, etc.)

2.2.2 Dataset Overview (10 points)

- Total number of observations (rows)
- Total number of variables (columns)
- Brief description of each variable including:
 - Variable name
 - Variable type (continuous, discrete, categorical nominal, categorical ordinal)
 - Units of measurement (if applicable)
 - Possible range or categories

Create a table summarizing all variables. Example format:

Variable	Type	Description	Range/Categories
age	Continuous	Patient age at enrollment	18-85 years
sex	Nominal	Biological sex	Male, Female
bp_category	Ordinal	Blood pressure category	Normal, Elevated, High

2.2.3 Missing Data Assessment (5 points)

- Which variables have missing data?
- What percentage of observations are missing for each variable?
- How will you handle missing data in your analysis?

2.2.4 Why This Dataset? (Bonus question, not graded)

In 2-3 sentences, explain what interested your group about this dataset and why it's important for public health or biomedicine.

3 Part C: Research Questions and Hypotheses (25 points)

Develop 2-3 specific, testable research questions based on your dataset. Each research question should include:

3.1 Research Question Format

For EACH research question (2-3 total), provide:

3.1.1 Question Statement (5 points per question)

State your research question clearly. Good research questions are:

- **Specific:** Focus on particular variables
- **Testable:** Can be answered with statistical methods
- **Relevant:** Important for public health or clinical practice

Examples of good research questions:

- “Is there a significant difference in mean systolic blood pressure between smokers and non-smokers?”
- “Are cholesterol levels normally distributed in this population?”
- “Is there an association between BMI category and diabetes diagnosis?”

Examples of poor research questions:

- “What can we learn from this data?” (too vague)
- “How does everything relate to everything?” (not specific)
- “What causes heart disease?” (too broad, requires complex modeling)

3.1.2 Formal Hypotheses (5 points per question)

For each research question, state:

- **Null hypothesis** (H_0): Statement of no effect/no difference
- **Alternative hypothesis** (H_A or H_1): What you expect to find

Use proper statistical notation where appropriate.

Example:

- H_0 : $\mu_{\text{smokers}} = \mu_{\text{non-smokers}}$ (no difference in mean BP)
- H_A : $\mu_{\text{smokers}} \neq \mu_{\text{non-smokers}}$ (two-sided alternative)

3.1.3 Public Health Significance (5 points per question)

For each research question, explain:

- Why is this question important?
- What are the potential implications if your hypothesis is supported?
- How might results inform public health policy or clinical practice?

Limit to 3-5 sentences per question.

3.1.4 Planned Statistical Method (5 points per question)

For each research question, specify:

- Which statistical test/method you will use (based on weeks 1-8 content)
- Why this method is appropriate for your variables and question
- What assumptions you need to check

Methods you have learned (weeks 1-8):

- Descriptive statistics and visualization
- One-sample t-test
- Two-sample t-test (independent samples)
- Paired t-test
- One-way ANOVA
- Nonparametric tests (Mann-Whitney U, Kruskal-Wallis, Wilcoxon signed-rank)
- Correlation (Pearson, Spearman)

4 Part D: Initial Data Exploration (25 points)

Conduct a thorough exploratory data analysis (EDA) of your dataset. This section should include:

4.1 Summary Statistics (10 points)

For your key variables of interest:

4.1.1 Continuous Variables

Provide:

- Sample size (n)
- Mean and median
- Standard deviation and IQR
- Minimum and maximum values
- Any outliers identified

Create summary tables. Example:

Variable	n	Mean	SD	Median	IQR	Range
Age (years)	500	52.3	12.1	51.0	44-60	18-85
SBP (mmHg)	500	128.5	18.2	125.0	115-140	90-195

4.1.2 Categorical Variables

Provide:

- Frequency counts for each category
- Percentages
- Mode (most common category)

Create frequency tables. Example:

Category	Count	Percentage
Male	245	49.0%
Female	255	51.0%
Total	500	100.0%

4.2 Data Visualization (10 points)

Create appropriate visualizations for your key variables. Include AT LEAST:

- **2-3 histograms** for continuous variables
 - Assess distribution shape (normal, skewed, bimodal?)
 - Include reference lines for mean/median if helpful
- **2-3 box plots** comparing groups
 - Show distribution differences between categories
 - Identify outliers visually
- **1-2 bar charts** for categorical variables
- **1-2 scatter plots** if examining relationships between continuous variables

All plots must:

- Have clear titles
- Have labeled axes with units
- Be large enough to read easily
- Be generated in R (no screenshots from other software)

4.3 Distribution Assessment (5 points)

For each continuous variable you plan to analyze:

- Assess normality visually (histogram, Q-Q plot)
- Comment on skewness or outliers
- Discuss whether parametric tests are appropriate or if you need nonparametric alternatives

Example discussion:

“Age appears approximately normally distributed based on the histogram, with no extreme outliers. The Q-Q plot shows points falling close to the reference line. Therefore, parametric methods (t-tests, ANOVA) are appropriate for this variable.

Systolic blood pressure shows slight right skewness with 3 potential outliers above 180 mmHg. We will verify these are not data entry errors. If skewness persists, we may use Mann-Whitney U test instead of two-sample t-test.”

5 Part E: Preliminary Analysis Plan (15 points)

5.1 Statistical Methods Selection (8 points)

For each research question from Part C, describe in detail:

1. **Which test you will use** (e.g., two-sample t-test, one-way ANOVA, Mann-Whitney U)
2. **Why this test is appropriate**
 - What are your independent and dependent variables?
 - What type of data do you have?
 - Does the test match your research question?
3. **Assumptions to check**
 - Independence of observations
 - Normality (if parametric)
 - Equal variances (if required)
 - Sample size requirements
4. **How you will check assumptions**
 - Visual methods (histograms, Q-Q plots, box plots)
 - Statistical tests (Shapiro-Wilk, Levene's test)
 - What you will do if assumptions are violated

5.2 Potential Challenges and Solutions (4 points)

Identify potential issues you foresee and how you plan to address them:

- **Missing data:** How much is missing? Will you use complete case analysis, imputation, or another approach?
- **Outliers:** How will you handle extreme values? Remove? Transform? Use robust methods?
- **Small sample sizes:** Do you have adequate power for your planned tests?
- **Assumption violations:** What alternatives will you use if parametric assumptions fail?
- **Multiple comparisons:** Are you conducting multiple tests? Do you need to adjust p-values?

5.3 Analysis Timeline (3 points)

Create a brief timeline for completing your final project:

Week	Tasks
Week 8	Complete data exploration (THIS ASSIGNMENT)
Week 9	Learn contingency tables/chi-square; apply to dataset if relevant
Week 9	Conduct hypothesis tests; create results tables
Week 10	Learn regression; apply to dataset if relevant
Week 10	Finalize analysis; create visualizations
Week 10	Write final report; create presentation slides
Week 10 (end)	Practice presentation; submit all materials

6 Part F: R Code and Reproducibility (Appendix)

6.1 Code Organization

Submit well-organized, fully commented R code that performs all analyses described in Parts B-E.

6.1.1 Required Code Sections

Your R code must include:

1. Header Comments

```
1 # H524 Project Preparation - Group [Name]
2 # Date: [Date]
3 # Group Members: [Names]
4 # Dataset: [Dataset name]
5 # Purpose: Exploratory data analysis and research question development
```

2. Package Loading

```
1 # Load required packages
2 library(tidyverse) # For data manipulation and visualization
3 library(here)      # For file path management
4 # Add other packages as needed
```

3. Data Import and Cleaning

```
1 # Import dataset
2 data <- read.csv("your_dataset.csv")
3
4 # Check structure
5 str(data)
6 summary(data)
7
8 # Handle missing data
9 # [Your code here]
10
11 # Check for duplicates
12 # [Your code here]
13
14 # Data type conversions if needed
15 # [Your code here]
```

4. Exploratory Data Analysis

```
1 # Summary statistics for continuous variables
2 # [Your code here]
3
4 # Frequency tables for categorical variables
5 # [Your code here]
```

```

6
7 # Create visualizations
8 # Histograms
9 # Box plots
10 # Scatter plots
11 # [Your code here]

```

5. Assumption Checking

```

1 # Normality assessment
2 # Q-Q plots
3 # Shapiro-Wilk test
4 # [Your code here]
5
6 # Check for outliers
7 # [Your code here]
8
9 # Variance homogeneity (if needed)
10 # [Your code here]

```

6. Preliminary Statistical Tests (optional at this stage)

```

1 # If you want to run preliminary tests, include them here
2 # Document results
3 # [Your code here]

```

6.2 Code Quality Requirements

Your code will be evaluated on:

- **Reproducibility (5 points)**
 - Can the instructor run your code without errors?
 - Are file paths handled properly? (Use relative paths or `here()` package)
 - Are all required packages listed?
 - Does the code produce all tables and figures in your report?
- **Organization and Clarity (5 points)**
 - Is code well-organized into logical sections?
 - Are variable names meaningful?
 - Is there adequate white space?
- **Documentation (5 points)**
 - Are there comments explaining what each section does?
 - Are complex operations explained?
 - Would another student understand your code?

6.3 RMarkdown Template (RECOMMENDED)

We strongly recommend using RMarkdown to integrate your code and written report. A template is provided below:

```
1 ---
2 title: "H524_Project_Preparation"
3 author: "Group_Name:[Your_names]"
4 date: "'r_Sys.Date()'"
5 output:
6   pdf_document:
7     toc: true
8     number_sections: true
9 ---
10
11 # Part A: Group Information
12
13 [Your content here]
14
15 # Part B: Dataset Description
16
17 [Your content here]
18
19 ```{r setup, include=FALSE}
20 knitr::opts_chunk$set(echo = TRUE, warning = FALSE, message = FALSE)
21 library(tidyverse)
22 ```
23
24 ```{r load-data}
25 # Load your dataset
26 data <- read.csv("your_dataset.csv")
27 ```
28
29 # Part C: Research Questions
30
31 [Your content here]
32
33 # Part D: Data Exploration
34
35 ## Summary Statistics
36
37 ```{r summary-stats}
38 # Your summary statistics code
39 ```
40
41 ## Visualizations
42
43 ```{r histograms}
44 # Create histograms
45 ```
46
47 # Part E: Analysis Plan
48
49 [Your content here]
```

```
50  
51 # Appendix: Full R Code  
52  
53 ```{r ref.label=knitr::all_labels(), echo=TRUE, eval=FALSE}  
54 # This automatically includes all code chunks  
55 ```
```

Grading Rubric (100 points total)

Component	Points	Key Criteria
Part A: Group Organization	15	
Team information	5	Complete roster with clear roles
Collaboration plan	5	Concrete meeting/communication plan
Individual contributions	5	Each member provides honest statement
Part B: Dataset Description	20	
Source and context	5	Complete citation and background
Variable documentation	10	Comprehensive table with all variables
Missing data assessment	5	Identification and handling plan
Part C: Research Questions	25	
Question clarity	5	Specific, testable, relevant (per question)
Hypotheses formulation	5	Proper null/alternative statements
Public health significance	5	Clear justification of importance
Method selection	5	Appropriate test chosen with rationale
Part D: Data Exploration	25	
Summary statistics	10	Complete tables for key variables
Visualizations	10	Multiple appropriate plots with labels
Distribution assessment	5	Thoughtful evaluation of assumptions
Part E: Analysis Plan	15	
Methods justification	8	Detailed plan with assumption checks
Challenges identified	4	Realistic issues with solutions
Timeline	3	Feasible schedule for completion
Part F: R Code Quality	15	
Reproducibility	5	Code runs without errors
Organization	5	Clear structure, good naming
Documentation	5	Adequate comments throughout
Professional Presentation	-5	Deductions for poor formatting, typos
TOTAL	100	

Additional Evaluation Notes

- **Group work expectations:** All members should contribute meaningfully. Individual contribution statements will be reviewed.
- **Statistical rigor:** Methods must be appropriate for your data and research questions.
- **Clarity of writing:** Scientific writing should be clear, concise, and professional.
- **Reproducibility:** Your code must run on the instructor's computer with only minor path changes.

AI Usage Guidelines

AI-Enhanced Learning for Group Projects

This is a GROUP assignment where collaboration and learning are paramount. AI tools can assist your learning, but you must understand and verify everything you submit.

Encouraged AI Uses:

- + **Code debugging:** “Why is this R code giving an error?”
- + **Syntax help:** “How do I create a grouped box plot in ggplot2?”
- + **Concept clarification:** “Explain the difference between parametric and nonparametric tests”
- + **Writing improvement:** “Help me make this explanation clearer” (after you write a draft)
- + **Method selection:** “I have continuous outcome and 3 groups. What test should I use?”

Prohibited AI Uses:

- **Complete delegation:** Asking AI to write your entire analysis
- **Hypothesis generation:** AI should not choose your research questions
- **Interpretation without understanding:** Copying AI explanations you don’t comprehend
- **Skipping group discussion:** Using AI instead of discussing with teammates

Required AI Documentation:

If you use AI tools significantly, include a brief section at the end of your report:

- Which AI tool(s) did you use? (e.g., ChatGPT, Claude, GitHub Copilot)
- For what purposes? (e.g., debugging code, explaining statistical concepts)
- How did you verify the AI’s suggestions?
- What did you learn from the AI interaction?

Remember:

- You are responsible for all content you submit
- You must understand every line of code and every statistical interpretation
- AI can assist learning, but cannot replace learning
- The goal is to develop YOUR group’s analytical skills

Submission Requirements

What to Submit (One submission per group)

1. Main Report (PDF)

- Generated from RMarkdown (strongly recommended) OR
- Word document converted to PDF OR
- LaTeX-generated PDF
- Must include all Parts A-E
- Must include all tables, figures, and narrative text
- Expected length: 8-12 pages (not including code appendix)

2. R Code File(s)

- Submit .Rmd file (if using RMarkdown) OR
- Submit .R script file with all code
- Ensure code is reproducible
- Include comments throughout

3. Dataset File

- If using a curated dataset: Include citation/link (no need to upload)
- If using custom dataset: Upload data file (CSV format preferred)
- Include data dictionary if not already documented

4. Individual Contribution Statements

- Can be included at end of main PDF OR
- Submit as separate document
- Each group member must provide their own statement

File Naming Convention

Use clear, consistent file names:

- Main report: `H524_ProjectPrep_GroupName.pdf`
- R code: `H524_ProjectPrep_GroupName.R` or `.Rmd`
- Dataset: `H524_ProjectPrep_GroupName_data.csv`

Canvas Submission

- ONE group member submits all files
- All other group members should be listed in the Canvas submission
- Submit by 11:59 PM on the due date (end of Week 8)
- Late penalty: 10% per day (unless prior arrangement made)

Example Research Question Development

To help you develop strong research questions, here are examples using a hypothetical cardiovascular disease dataset:

Example 1: Comparing Two Groups

Research Question: Is there a significant difference in mean systolic blood pressure between patients with and without diabetes?

Hypotheses:

- H_0 : $\mu_{\text{diabetes}} = \mu_{\text{no diabetes}}$ (no difference in mean SBP)
- H_A : $\mu_{\text{diabetes}} \neq \mu_{\text{no diabetes}}$ (there is a difference)

Variables:

- Independent: Diabetes status (categorical: yes/no)
- Dependent: Systolic blood pressure (continuous, mmHg)

Statistical Method: Two-sample t-test (or Mann-Whitney U if SBP not normally distributed)

Assumptions to Check:

- Independence: Patients are independent observations
- Normality: Check histogram and Q-Q plot for SBP in each group
- Equal variances: Levene's test or examine box plots

Public Health Significance: Understanding blood pressure differences by diabetes status can inform screening recommendations and highlight the need for intensive BP management in diabetic patients.

Example 2: Comparing Multiple Groups

Research Question: Do mean cholesterol levels differ significantly across three BMI categories (normal weight, overweight, obese)?

Hypotheses:

- H_0 : $\mu_{\text{normal}} = \mu_{\text{overweight}} = \mu_{\text{obese}}$ (all group means equal)
- H_A : At least one group mean differs from the others

Variables:

- Independent: BMI category (categorical: 3 levels)
- Dependent: Total cholesterol (continuous, mg/dL)

Statistical Method: One-way ANOVA (or Kruskal-Wallis if assumptions violated)

Assumptions to Check:

- Independence: Patients are independent
- Normality: Check histograms for cholesterol in each BMI group
- Equal variances: Levene's test; examine box plots

Public Health Significance: If higher BMI is associated with elevated cholesterol, this supports weight management interventions for cardiovascular disease prevention.

Example 3: Correlation/Association

Research Question: Is there a significant correlation between age and resting heart rate in this population?

Hypotheses:

- H_0 : $\rho = 0$ (no correlation between age and heart rate)
- H_A : $\rho \neq 0$ (there is a correlation)

Variables:

- Variable 1: Age (continuous, years)
- Variable 2: Heart rate (continuous, beats per minute)

Statistical Method: Pearson correlation (if both variables normally distributed) or Spearman correlation (if not)

Assumptions to Check:

- Linearity: Create scatter plot
- Normality: Check histograms for both variables
- Outliers: Identify extreme values that might influence correlation

Public Health Significance: Understanding age-related changes in heart rate helps establish normal ranges and identify patients with abnormal cardiovascular responses.

Tips for Your Group

- Start with simple, clear questions
- Make sure your questions can be answered with methods from weeks 1-8
- Choose questions that genuinely interest your group
- Consider clinical or public health relevance
- Ensure your dataset has the necessary variables

Required R Packages

Your analysis will likely use the following R packages. Install them before beginning:

```
1 # Core packages
2 install.packages("tidyverse") # Data manipulation and visualization
3 install.packages("here")      # File path management
4
5 # Summary statistics
6 install.packages("psych")     # describe() function
7 install.packages("Hmisc")     # More descriptive stats
8
9 # Visualization
10 install.packages("ggplot2")   # Already in tidyverse, but can load separately
11 install.packages("gridExtra") # Arrange multiple plots
12
13 # Statistical tests
14 # Base R includes t.test(), wilcox.test(), aov(), chisq.test(), cor.test()
15
16 # Assumption checking
17 install.packages("car")       # Levene's test, Q-Q plots
18
19 # RMarkdown (if using)
20 install.packages("rmarkdown")
21 install.packages("knitr")
```

Getting Help

If your group encounters difficulties:

Office Hours

- Instructor office hours: [Days/times listed on syllabus]
- TA office hours: [Days/times listed on syllabus]
- Special project office hours Week 7.5 for dataset approval/questions

Canvas Discussion Board

- Post general questions (not dataset-specific answers)
- Help other groups when you can
- Instructor monitors regularly

Email

- For private group concerns
- For dataset approval requests
- For deadline extensions (must request BEFORE due date)

Resources

- Course textbook: Pagano & Gauvreau, *Principles of Biostatistics*
- R documentation: <https://www.r-project.org/>
- RMarkdown guide: <https://rmarkdown.rstudio.com/>
- Course Canvas page: lecture notes, example code

Academic Integrity

Collaboration Policy:

- **Within your group:** Full collaboration is expected and encouraged
- **Between groups:** You may discuss general concepts, but not share code or specific analyses
- **Outside resources:** You may consult textbooks, online tutorials, AI tools, but must cite sources
- **Plagiarism:** Copying from other groups, previous years, or uncited sources is prohibited

What this means:

- It's OK to say: "Our group is using ANOVA for our main analysis. What are you using?"
- It's NOT OK to say: "Can I see your R code for the ANOVA?"
- It's OK to use AI to debug code or explain concepts
- It's NOT OK to have AI write your entire analysis

All group members are responsible for ensuring academic integrity. Violations will result in zero points for the assignment and possible further consequences per university policy.

Good luck with your project!

*This is an opportunity to apply your biostatistics knowledge to real-world data.
Ask questions, collaborate with your team, and enjoy the process of discovery.*